

WAVES: AN INTERACTIVE TUTORIAL



Kyle Forinash and Wolfgang Christian

Waves: An Interactive Tutorial (Forinash and Christian)

Kyle Forinash and Wolfgang Christian

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Preface

Waves: An Interactive Tutorial is a set of 33 exercises designed to teach the fundamentals of wave dynamics. It starts with very simple wave properties and ends with an examination of nonlinear wave behavior. The emphasis here is on the properties of waves which are difficult to illustrate in a static textbook figure. Simulations are not a substitute for laboratory work. However they allow for visualization of processes that cannot normally be seen (for example electric and magnetic fields). They allow for visualization of process that are too fast (for example waves) to follow in real time or too small to see (for example thermodynamics at the molecular scale). They allow manipulation of processes which might be dangerous (collisions) or hard to experiment with (waves). They also allow for easy repetition. For all of these reasons, simulations are an excellent way to introduce students to the complex phenomena of waves.

The tutorial may be used in conjunction with a text or as a stand-alone introduction to waves. Exposure to calculus and basic physics is assumed in the latter sections. The material was constructed in such a way that the student is required to actively work their way through the material instead of passively reading printed text at a low comprehension level. The objective is to replace the traditional textbook for the course (but not necessarily replace the entire course) with computer aided instruction that requires active learning on the part of the student. The goal was to create a guided text that integrates the strengths of printed, static textbooks and the interactive dynamics of the Internet to engage the student in actively learning the physics of waves.

Play the above simulation and observe the propagation of the wave along a beaded string. How can this motion be expressed mathematically? What would happen if the string is attached to a wall? What happens if the mass of the beads is increased? We will develop answers to these and related questions in this tutorial but what is as important is the mental image gained from a visualization of the phenomena.

Interactive Engagement

We have known for sometime that active teaching methods work better than lecturing in the classroom. (See R.R. Hake, "Interactive-engagement vs traditional methods: A six-thousand-student survey of mechanics test data for introductory physics courses", *Am. J. Phys.*, Jan. (1998)). Many instructors now use collaborative group work, lecture demonstration, Socratic dialog and other interactive techniques in their classroom. But what about the textbook? Students now have access to electronic textbooks for sale or rent but these are still mostly static (pdf) files. All that has been done is to speed up access to the same material that existed in hard copy. Except for the introduction of color pictures, not much has changed since Guttenberg's time.

Physics textbooks are notoriously hard to read. In Student Evaluation of Teaching (SET) score results, the perception of the physics textbook ALWAYS receives the lowest score, regardless of which text is chosen or who teaches the class. When (if?) students read the text, what do they get out of it? Do they actively engage in the subject or do they just look at the words and pictures? Some research evidence hints that, at least for physics, students don't get much from a traditional textbook but rather depend on the instructor to interpret for them because they do not know how to read a physics book. But what if the textbook required answering questions and manipulating computer simulations as the student worked through a topic? The goal of this tutorial is to explore the possibility that a textbook which required the student to be engaged would lead to better student understanding of the material; to make a better textbook, an *active learning textbook*.

In the education world the Internet and course management technology is still too often used as a one way communication tool, simply making it a bit more convenient for students in a conventional course to receive the same or similar material which previously was handed out in class. But the same technology can be used as a two way communication tool where students are actively engaged in manipulating course material, controlling both simulations and live experiments, collaborating with other students and interacting with the instructor. Many of these activities can be done in a laboratory setting as an integral part of a regularly scheduled course, performed asynchronously as homework assignments or completed as a component of a new hybrid kind of distance learning course.

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CHAPTER OVERVIEW

1: Basic Properties

The fundamental properties of waves, such as frequency, amplitude, wavelength, and motion direction, are quite different from those of rigid objects studied in mechanics. These tutorials explore the basics of wave motion.

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1.1: Sine Wave

Tutorial 1.1: Sine Waves

Imagine a perfect, smooth wave out on the ocean far enough from shore so that it has not started to break (complications involved in describing real waves will be discussed later in this tutorial). If we take a snapshot picture of this wave at a single instant in time and measure the distance in meters from one peak to the next we have measured the **wavelength**, λ of the wave.

If instead we watch a floating cork at a single location in space and measure the time in seconds between arriving peaks we have measured the **period**, T of the wave. We could also measure the number of times the cork bobs up, down and back up per second which would be the **frequency** in hertz or cycles per second. The period and frequency are inverses of each other: $f = 1/T$.

The height of the wave at any location and time, measured from the middle, or equilibrium position is the **displacement**. The maximum displacement is called the **amplitude**.

As a first approximation, water waves, electromagnetic waves and many other kinds of waves can be modeled by the mathematical functions sine or cosine or some combination of them. For a wave traveling along the x axis the mathematical description of the displacement of a wave at location x and time t can be written as $y(x, t) = A \sin(kx - \omega t + \varphi)$ where A is the **amplitude** (maximum height measured from the middle of the wave). Here we have used the **wavenumber** $k = 2\pi/\lambda$ (measured in radians per meter), the **angular frequency**, $\omega = 2\pi f$ (measured in radians per sec), and φ , the **phase** angle (in radians) which are often easier to use mathematically.

In this and all of the following exercises the angles will be in radians (Hint: Set your calculator to radian mode to avoid problems later on!)

With the following physlet you can explore different values of amplitude, wavenumber, wavelength, angular frequency, frequency, period and phase. Make any changes you wish in the $y(x, t) =$ window, hit enter or return and then the play button, ▶, to see the wave in action. The initial values are amplitude, $A = 1.2$ m; wavenumber, $k = 2.0$ rad/m; angular frequency, $\omega = 0.8$ rad/s; and phase angle, $\varphi = 10.0$ radians. Clicking in on the graph shows the coordinates of the cursor in a yellow box in the lower left corner.

Sine Wave Simulation

Questions:

Exercise 1.1.1

Stop the simulation, double the maximum amplitude (from 1.2 m to 2.4 m) enter and play. What effect does maximum amplitude, A , have on the wave?

Exercise 1.1.2

Stop the simulation, double the wave number, k , (from 2.0 rad/m to 4.0 rad/m) enter and play. What effect does the wave number have on the wave?

Exercise 1.1.3

Stop the simulation, double the angular frequency, ω , (from 0.8 rad/s to 1.6 rad/s) enter and play. What effect does the angular frequency have on the wave?

Exercise 1.1.4

Stop the simulation, double the phase, φ , (from 10.0 to 20.0 radians) enter and play. Try several different values for the phase. What effect does the phase have on the wave?

Exercise 1.1.5

Go back to the original wave by clicking the reload button, ↺. Pause the wave and measure the wavelength, λ , on the graph (find the x location of two successive peaks or troughs using the cursor; the wavelength is the x distance between peaks or

troughs). Calculate the wavenumber, k , from this wavelength. How does your value for wavenumber compare with the wavenumber in the equation?

Exercise 1.1.6

Now start the original wave in motion with the play button. Use the time numbers in the lower panel to find the period, T , of the wave (the time from when one peak passes a point until the next peak passes the same point). To get an accurate number you can use the step buttons. From the period you measure, calculate the angular frequency, ω . How does your value for angular frequency compare with the angular frequency in the equation?

Exercise 1.1.7

Go back to the original wave using the reload button. Change the minus sign in the equation between kx and ωt to a plus sign and click 'play'. What does changing this sign do to the wave?

Exercise 1.1.8

Now change the plus sign in front of the phase to a minus sign, enter and play. Try several values of phase (you may want to use the pause button to be sure you can tell what is happening). What does changing this sign do to the wave?

Exercise 1.1.9

In general the power transmitted by a wave, measured in watts, is proportional to the amplitude squared. What happens to the power if the amplitude is doubled? What happens to the power if the amplitude is cut in half?

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1.2: Speed of a Wave

Tutorial 1.2: Speed of a Wave

There are three different velocities involved with describing a wave, one of which will be introduced here. The **velocity** of the wave, v , is a constant determined by the properties of the medium in which the wave is moving. The velocity is a vector which gives the forward **speed** of the wave and the direction the wave is traveling. For now we will not worry about direction since the waves being discussed will all be assumed to travel along the x -axis. The speed of a sine wave is given by $v = \lambda/T$ in meters per second where wavelength and period for a sine wave were defined in the previous exercise.

In this simulation the original wave will remain in the window so that as you make changes to $f(x, t)$ you can see how the new wave (in red) compares to the original ($g(x, t)$, in blue).

Wave Speed Simulation

Questions:

Exercise 1.2.1

Determine the speed of the wave in the simulation using $v = \lambda/T$ where wavelength and period are determined from the simulation as you did in the previous exercise using the mouse to find the wavelength and the time to find the period. What is the forward speed of this wave?

Exercise 1.2.2

The speed of this wave is also given mathematically by $v = \omega/k$ since $\omega = 2\pi f = 2\pi/T$ and $k = 2\pi/\lambda$. What is the speed of this wave based on the values of ω and k in the equation? Does this match the speed you got from the simulation?

Exercise 1.2.3

Reload the initial conditions and experiment with values of the wavenumber both smaller and larger than 2.0 rad/m keeping the angular frequency fixed. How does the wavenumber change the speed of the wave?

Exercise 1.2.4

Reload the initial conditions and experiment with values of the angular frequency both smaller and larger than 0.8 rad/s keeping the wavenumber fixed. How does the angular frequency change the speed of the wave?

This simulation is *misleading* in one important way. In the simulation you can set any combination of angular frequency and wavenumber you choose and so have any speed you want for the wave. But for mechanical and acoustic waves the speed is determined by the medium in which the wave travels. As we will see, for these waves it is often the case that $v = \omega/k$ so that the angular frequency and wavenumber are inversely proportional with v a constant. Examples:

- For sound waves in a fluid (for example air) the speed is determined by $v = (B/\rho)^{1/2}$ where B is the bulk modulus or compressibility of the fluid in newtons per meter squared and ρ is the density in kilograms per cubic meter.
- For waves in a solid the speed is determined by $v = (Y/\rho)^{1/2}$ where Y is Young's modulus or stiffness in newtons per meter squared and ρ is the density in kilograms per meter cubed.
- For waves on a string the speed is determined by $v = (T/\mu)^{1/2}$ where T is the tension in the string in newtons and μ is the mass per length in kilograms per meter.
- Although electromagnetic waves do not need a medium to travel (they can travel through a vacuum) their speed in a vacuum, $c = (1/\mu_0\epsilon_0)^{1/2}$ is governed by two physical constants, the permeability μ_0 and the permittivity, ϵ_0 .
- If an electromagnetic wave enters a medium (such as light going through air, or water or glass) the speed changes because the permittivity and/or permeability may be different. The amount the speed changes is given by the **index of refraction** $n = c/v$ where c is the speed of light and v is the speed in the medium.

As we will see later, it is the case that speed can sometimes depend on the frequency of the wave, a phenomenon known as dispersion.

Exercise 1.2.5

Does sound travel slightly faster on a hot day or a cool day? Does sound travel faster or slower if the humidity is high?

Exercise 1.2.6

Density is relatively easy to measure. What would be a clever way to measure Young's modulus, Y for a solid?

Exercise 1.2.7

Do waves on a string travel faster or slower if the string is tighter? Do waves on a string travel faster or slower if the string is thicker?

Exercise 1.2.8

In general the index of refraction is larger than one. What does this tell you about the speed of light in glass?

Exercise 1.2.9

Reload the initial conditions with the 'reload' button. For a wavenumber of 4.0 rad/m experiment to find the correct angular frequency which gives the original speed of the wave you found in Exercises 1.2.1 and 1.2.2 (you should be able to see from the simulation when the new wave is traveling at the same speed as the original).

Exercise 1.2.10

Calculate the wavenumber which gives the speed of the original wave for angular frequencies of 0.4 , 0.6 , 1.0 , and 1.2 rad/s using the relationship in Exercise 1.2.2. Check your answers with the simulation if you are in doubt.

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1.3: Transverse Waves

Tutorial 1.3: Transverse Waves

Transverse waves are the kind of wave you usually think of when you think of a wave. The motion of the material constituting the wave is up and down so that as the wave moves forward the material moves perpendicular (or **transverse**) to the direction the wave moves. Examples of transverse waves include waves on a string and electromagnetic waves. Water waves can be approximately transverse in some cases.

The following simulation shows a graph of the motion of one location, the red circle, on a string which has a transverse wave on it. The vertical location of points on the string (represented by the circles) as a function of horizontal location along the x-axis and time is again described mathematically by $y(x, t) = A \sin(kx - \omega t + \varphi)$. Notice that, while the wave moves forward along the string, the red circle does not (in fact none of the circles move forward).

Transverse Wave

Questions:

Exercise 1.3.1

Play the animation (lower left button). From the graph, what are the amplitude and period of the motion of the red dot?

Exercise 1.3.2

Clicking on the lower panel gives the mouse location (in the yellow box) which in this case are the x and y location of points on the wave. Use these numbers to determine the wavelength of the wave (this is easiest to do with the animation paused or finished).

Exercise 1.3.3

From the period and wavelength that you just measured, calculate the forward speed of the wave (as you did in the previous simulation).

A *second* velocity associated with a wave is how fast the material of the wave moves up and down at a single location (the vertical speed of the circles in the simulation). This velocity, the **transverse velocity**, is not a constant but is a function of location and time (different places on the wave move upward or downward at different speeds at different times). Since velocity is the rate of change of position, this second speed (in the y direction) is given by the derivative of the amplitude with respect to time:

$$v(x, t) = \partial y(x, t) / \partial t = -A\omega \cos(kx - \omega t + \varphi)$$

Notice that the maximum speed of a section of the wave at location x and time t will be given by $v_{\max} = A\omega$. We use a partial derivative here because $y(x, t)$ is a function of two variables.

Exercise 1.3.4

Click on the 'Velocity' button and then 'play'. The upper graph now gives the speed of the red circle in the y -direction as a function of time. What is the maximum speed (approximately) of the red circle according to the graph? How does this compare with the speed of the wave which you found in 1.3.3; are they the same or different?

Exercise 1.3.5

Where is the red dot (relative to the rest position before the wave passes) when the maximum transverse velocity occurs? Where is it when the transverse velocity is approximately zero?

Exercise 1.3.6

Using $v_{\max}$ from the graph, the amplitude from 1.3.1 and $v_{\max} = A\omega$, what is the angular frequency? How does this compare with the value calculated from the period?

Since points on the wave change their transverse velocity over time there must also be a vertical or **transverse acceleration**. Since acceleration is the time rate of change of velocity we have $a(x, t) = \partial v(x, t) / \partial t = -A\omega^2 \sin(kx - \omega t + \varphi)$ where the **maximum acceleration** is $a_{\max} = A\omega^2$.

Exercise 1.3.7

Calculate the maximum acceleration of the red circle. What are the units of this acceleration if amplitude is in meters and angular frequency is in radians per second?

Exercise 1.3.8

Based on the equation for acceleration, where will the red circle be when the acceleration is a maximum? Where will it be when the acceleration is approximately zero?

Exercise 1.3.9

Carefully state the relationship between position, velocity and acceleration. When the position is zero (equilibrium position of the red dot) what are the velocity and acceleration? When the position is a maximum, what are the velocity and acceleration?

Exercise 1.3.10

State in your own words the difference between wave speed and transverse speed of a wave.

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1.4: Simple Harmonic Motion

Tutorial 1.4: Simple Harmonic Motion I

The following is a simulation of a mass on a spring. The graph shows the y location of the mass. The force acting on the mass in this case is called a **Hooke's Law** force: $F = -\kappa y$ where κ is called the **spring constant**, in N/m indicating the stiffness of the spring and y is the location of the mass from some equilibrium position. (Careful! κ is **not** the same as the wavenumber, $k = 2\pi/\lambda$.)

Simple Harmonic Motion

Questions:

Exercise 1.4.1

Drag the mass to some initial location using the mouse. Click 'play' button (lower left) to see the motion and the graph of the mass location. How is the graph you see here similar to the graph you saw for the motion of the red circle in the transverse wave case?

Exercise 1.4.2

Clicking on the graph shows the coordinates of the mouse in a yellow box at the lower left. Determine the period, frequency and angular frequency of this motion from the values on the graph (Hint: Frequency in Hz is the inverse of period; angular frequency is $\omega = 2\pi f$).

Exercise 1.4.3

Check the 'show velocity' box to see graphs of both position and velocity. Where is the mass when the velocity is a maximum? Where is the mass when the velocity becomes zero?

Exercise 1.4.4

Try different spring constant values, κ , between 0.5 N/m to 5.0 N/m, resetting and releasing the mass at the same point each time. What is the relationship between spring constant and frequency?

Exercise 1.4.5

For the same spring constant try releasing the ball at several different amplitudes. What is the relationship between amplitude and frequency?

For a given spring the frequency is determined by the mass hanging on it and the stiffness of the spring; $f = (\kappa/m)^{1/2}/2\pi$.

Exercise 1.4.6

Measure the frequency for the case of a spring constant, κ equal to 2.0. What must be the mass on the end of this spring? Verify that you get the same mass by measuring the frequency for several different spring constants (this is equivalent to hanging the same mass on several different springs).

Exercise 1.4.7

What happens to the period of oscillation of a spring-mass system if the mass is doubled?

In the previous simulation (1.3: Transverse Waves) the red circle moved up and down as the result of a transverse wave traveling horizontally along the string of particles. The equation for the motion of the entire string is $y(x, t) = A \sin(kx - \omega t + \varphi)$. If we were to assume that the red particle was located at $x = 0$ the equation describing the motion of just the red circle would be $y(t) = A \sin(-\omega t + \varphi)$ where A is the maximum amplitude, ω is the angular velocity and φ is the phase. This is also the formula for **simple harmonic motion** which describes the location of a mass on a spring as a function of time. In other words the motion of

each point on a transverse wave is exactly the same as if each of those points was undergoing simple harmonic motion but with a slightly different phase from its neighbor.

Exercise 1.4.8

Using $y(t) = A \sin(-\omega t + \varphi)$ and a calculator, what is the location of the mass when $t = 0$ and $\varphi = 0$? What is the location of the mass when $t = 0$ and $\varphi = \pi$ (don't forget we are in radian mode)? Explain what the phase angle, φ tells you about the initial position ($t = 0$) of the mass on the spring.

Exercise 1.4.9

If $y(t) = A \sin(-\omega t + \varphi)$ is the location of the mass on the spring and the time derivative of location is velocity, then the velocity of the mass is given by $v(t) = A\omega \cos(-\omega t + \varphi)$. Check the 'show velocity' box and run a simulation with an initial amplitude of 6.0 m and a spring constant of 2. From the graph find the angular frequency and calculate the maximum amplitude $v_{\max} = A\omega$. How does this number compare with the maximum value on the velocity graph; are they the same?

Exercise 1.4.10

Write down an expression for the acceleration of a mass on a spring, based on the position given by $y(t) = A \sin(-\omega t + \varphi)$ and the fact that acceleration is the second derivative of position with respect to time.

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1.5: Simple Harmonic Motion and Resonance

Tutorial 1.5: Simple Harmonic Motion and Resonance

The following simulation shows a driven, damped harmonic oscillator; a 1 kg mass on a spring with spring constant 2 N/m. The amplitude of the motion is graphed versus time. The initial position of the mass, x_0 , can be adjusted by dragging the mass to a starting position. The driving frequency, f , can be adjusted so we expect that for one particular frequency we will see the amplitude of the motion to be very large; in other words, resonance will occur.

Several other parameters can also be adjusted. b is the amount of friction in Ns/m (this could be air resistance or sliding friction or friction in the spring itself); v_0 in m/s is the initial velocity of the mass, and F_0 is the magnitude of the driving force in newtons.

In this (and many other simulations we will use) it is easier to write $\omega = 2\pi f$ where ω is the **angular frequency** in radians per second instead of having to write $2\pi f$ everywhere.

Simple Harmonic Motion II

Questions:

Exercise 1.5.1

Start with the default parameters, drag the ball to the maximum starting position (10 m) and hit play. How does this motion compare with simple harmonic motion (in the last chapter)?

Exercise 1.5.2

Reset the simulation, change the friction parameter, b to 0.5 Ns/m, drag the ball to the maximum starting position (10 m) and hit play. (You can also click the button at the top for under damped motion.) What happens? This motion is called **damped harmonic motion**.

Exercise 1.5.3

Try different values for b . How is the behavior of the mass for small values of b (less than one) different than for values larger than 2 Ns/m (be sure to use the same starting position each time)?

If the mass oscillates at least once before stopping the damping is called **under damped** motion. If the mass never quite gets back to the equilibrium position the motion is called **over damped**. The case where there is just enough damping so that an oscillation does not occur (the mass just barely makes it back to equilibrium) is called **critically damped** motion.

Exercise 1.5.4

Click the buttons on the top for under, over and critically damped motion. Describe the differences between these three cases.

Exercise 1.5.5

Reset the simulation, set the friction parameter, b to 1.0 Ns/m and the magnitude of the driving force, F_0 to 1.0 N. The angular frequency should be 1.0 rad/s also. Drag the ball to the maximum starting position and hit play. In this and the next few questions you will need to wait 10 s or so for the motion to stabilize. Describe the stable motion after the initial oscillations for this case. This motion is called **driven, damped harmonic motion**.

Exercise 1.5.6

Experiment with different amounts of force, keeping friction and angular frequency equal to one. Also start from the same position each time. What is the effect of larger values of force amplitude, F_0 on the final, stable motion of the mass? HINT: Use a screen shot of the graph to compare results.

Exercise 1.5.7

With damping, b set to 0.2 Ns/m and F_0 set to 1.0 N try several values of the angular driving frequency, ω . Start with a value of 1.1 rad/s and increase by 0.1 each time until you get to 1.8 rad/s. In each case, wait until the animation ends and measure the amplitude by clicking on the top of the curve near the end. The second number in the yellow box is the amplitude in m. Write down the amplitude for each driving frequency. Which driving frequency ended up giving the largest amplitude?

Resonance is defined to occur when a vibrating system is driven with a frequency that causes the largest amplitude. Generally this occurs when the driving frequency equals the natural frequency. An example of resonance is when a wine glass is driven by sound waves with a frequency equal to the natural frequency of the glass. If the amplitude becomes large enough the glass will break. There are many other examples of resonance. Musical instruments depend on the phenomena of resonance to produce fixed pitches, tuning a radio to a particular channel depends on selecting a resonance frequency and Magnetic Resonance Imaging (MRI) in the medical world uses resonance to form images of the inside of the human body.

As we saw in the previous chapter the **natural frequency**, written as f_0 is given by the stiffness of the spring, κ , and the mass; $f_0 = (\kappa/m)^{1/2}/2\pi$. In this simulation the mass is 1 kg and the spring constant is 2 N/m so $f_0 = 0.225$ Hz and the natural angular frequency, $\omega_0 = 2\pi f$ equals 1.41 rad/s. In the previous question you should have seen the maximum amplitude for a driving frequency of 1.41 rad/s. In other words a driving frequency, ω of 1.41 rad/s leads to resonance (maximum amplitude) because it equals the natural frequency, ω_0 .

Exercise 1.5.8

Use a calculator to find the natural frequency for a spring and mass system with $m = 2.0$ kg and $\kappa = 5.0$ N/m. What do you expect the resonance frequency to be for this case?

The mathematical formula that describes damped harmonic motion is $Ae^{-\gamma t} \cos(\omega t + \varphi)$ where $\gamma = b/2m$. Notice that this is the same cosine function for simple harmonic motion but the amplitude, A , is multiplied by an exponentially decreasing function of time, $e^{-\gamma t}$. So we expect the oscillation of a damped harmonic oscillator to be an up and down cosine function with an amplitude that decreases over time.

Exercise 1.5.9

Check to see if the formula $Ae^{-\gamma t} \cos(\omega t + \varphi)$ really does describe the behavior of damped harmonic motion in the simulation. To do this, use a graphing calculator (or go to [meta calculator](#) and chose graphing calculator or use [desmos calculator](#)) and plot $y = 10 * \exp(-.2 * x) * \cos(1. * x)$. This is the case $A = 10$ m; $b = 0.4$ Ns/m; $m = 1.0$ kg; and $\omega = 1.0$ rad/s. (You can cut and paste the equation into the online calculator). How does this graph compare with the simulation for these same parameters (note: you are only interested in the positive x values)?

The mathematical formula that describes driven, damped harmonic motion is $A_0 \cos(\omega t + \varphi)$ where $A_0 = (F_0/m)/((\omega^2 - \omega_0^2)^2 + 4\gamma^2\omega^2)^{1/2}$. In this case the amplitude, A_0 does not change over time but it is dependent on the driving frequency, ω .

Exercise 1.5.10

What happens to the amplitude, A_0 , when $\omega = \omega_0$ (assume all the other factors are constant numbers)? Is there any other combination of ω and ω_0 that gives a larger amplitude?

Exercise 1.5.11

Verify your previous answer by making a plot of $A_0 = (F_0/m)/((\omega^2 - \omega_0^2)^2 + 4\gamma^2\omega^2)^{1/2}$ for a range of driving frequency. To do this, use a graphing calculator to graph $y = 1.0/((1.4 \wedge 2 - x \wedge 2) \wedge 2 + 4 * .04 * 1.4 \wedge 2) \wedge .5$. Where does this graph have a maximum for positive values of x ?

Exercise 1.5.12

Based on the previous two questions, what is the resonance frequency of this system and how does this compare with the natural frequency?

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1.6: Longitudinal Waves

In comparing simulations on transverse waves (Tutorial 1.3) with vertical harmonic motion (Tutorial 1.4) we discovered that particles in a transverse wave move up with simple harmonic motion. In the previous exercise (Tutorial 1.5) we saw that harmonic motion can also occur in the horizontal direction. Can we also have a wave moving horizontally where the particles move with harmonic motion in the horizontal direction?

YES! **Longitudinal waves** are waves where the motion of the material in the wave is back and forth in the same direction that the wave moves. Sound waves (in air and in solids) are examples of longitudinal waves. When a tuning fork or stereo speaker vibrates it moves back and forth creating regions of compressed air (where the pressure is slightly higher) and regions in between where the air has a lower pressure (called a rarefaction). These compressions and rarefactions move out away from the tuning fork or speaker at the speed of sound. When they reach your ear they cause your eardrum to vibrate, sending signals through the rest of the ear to the brain.

Longitudinal waves can be described with the same mathematical functions as transverse waves: $y(x, t) = A \sin(kx - \omega t + \varphi)$ where now $y(x, t)$ is the *horizontal* (or longitudinal) displacement from equilibrium at location x and time t instead of the vertical displacement from equilibrium. As was the case for transverse waves the forward velocity of a longitudinal wave is given by $v = \lambda/T = \omega/k$.

The following simulation shows a graph of the longitudinal motion of one molecule, the red circle, in a collection of molecules which has a longitudinal wave passing through it, much like sound passing through air. A vertical line marks the equilibrium location of the red circle. Random thermal motions are not shown.

Longitudinal Waves

Questions:

Exercise 1.6.1

Click on 'Position' and then 'play'. Left clicking on the upper panel gives the time and amplitude of points on the graph in the yellow box. Do any of the circles travel all the way across the simulation to the other side? Explain.

Exercise 1.6.2

Left clicking on the upper panel gives the time and amplitude of points on the graph in the yellow box. Determine the maximum amplitude and the period of oscillation from the graph.

Exercise 1.6.3

Left clicking on the lower panel gives the x and y locations of points on the wave in a yellow box. Pause and step the animation until the red circle at its equilibrium position. Find the wavelength of the wave using the mouse by finding the distance between one place where the circles are clumped together to the next location (or from two successive locations where the circles are furthest apart). What is the wavelength?

Exercise 1.6.4

From the period and wavelength find the speed of this wave (Hint: The same equations work for both longitudinal and transverse waves).

For sound the frequency (inversely proportional to the wavelength) tells us something about the **pitch** of the sound. There are other aspects of pitch perception which involve other physical features of the wave but the main component of pitch is the frequency.

Exercise 1.6.5

What is the frequency of the wave in the simulation?

Exercise 1.6.6

Write an equation of the form $y(x, t) = A \sin(kx - \omega t + \varphi)$, filling in the values of A , k and ω for this wave. Assume the phase angle is zero.

Notice that the circles in the simulation move back and forth with a variable speed around an equilibrium position while the wave moves only in one direction with a constant speed. The velocity of the individual particles is given as before by the derivative of the amplitude: $v(x, t) = \partial y(x, t) / \partial t = -A\omega \cos(kx - \omega t + \varphi)$.

Exercise 1.6.7

Click on 'Velocity' and then 'play'. The upper graph now gives the velocity of the red circle as a function of time. What is the maximum velocity (approximately) of the red dot according to the graph? How does this compare with the velocity of the wave which you found in 1.6.4? How does it compare with $v_{\max} = A\omega$?

Exercise 1.6.8

In your own words, explain the difference between wave speed and particle velocity for a longitudinal wave.

Exercise 1.6.9

Where is the red dot relative to the vertical line when the maximum velocity occurs? Where is it when the velocity is approximately zero? What is the relationship between position and velocity.

Exercise 1.6.10

Take a derivative of velocity to find an expression for acceleration of particles in the material (the red dot). Show that the maximum acceleration is given by $a_{\max} = A\omega^2$.

Exercise 1.6.11

Calculate the maximum acceleration of the red dot using $a_{\max} = A\omega^2$. If amplitude is in meters and angular frequency in radians per second, what are the units of this acceleration?

As a sound wave moves through the air the air molecules do not move forward at the speed of sound but rather, oscillate back and forth as harmonic oscillators in the same general location while the sound wave passes (see question one). For sound waves the displacement amplitude (distance from the equilibrium location) tells us something about the pressure of the air at that location. Pressure is measured in pascals ($1 \text{ Pa} = 1 \text{ N/m}^2$) and pressure squared is proportional to the **intensity** of the sound wave, measured in W/m^2 .

The relationship between sound intensity, I measured in watts per meter squared and **loudness**, or sound intensity level (SIL) measured in decibels, is given by $SIL = 10 \log(I/I_0)$. Here $\log$ is the logarithm and $I_0 = 10^{-12} \text{ W/m}^2$ is a reference sound intensity at about the threshold of human hearing.

Exercise 1.6.12

For the following sound intensities, what is the equivalent SIL in decibels: Jet engine, 100 W/m^2 ; pain threshold, 1 W/m^2 ; vacuum cleaner, 10^{-4} W/m^2 ; conversation, 10^{-6} W/m^2 ; the rustle of leaves, 10^{-11} W/m^2 .

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1.7: Water Waves

Tutorial 1.7: Water Waves

Many real, physical waves are combinations of three kinds of wave motion; transverse, longitudinal and torsional (which we have not yet discussed). The following is a more accurate simulation of water waves (but it still does not show breaking wave behavior which will come later). The dots are locations of water molecules or small objects floating in the water.

Water Waves

Questions:

Exercise 1.7.1

What two types of wave motion are represented in the simulation?

Exercise 1.7.2

Are the wavelengths and periods the same for both types of motion (Hint: Use 'pause' and hold the left mouse button down to make measurements)?

Exercise 1.7.3

Determine the wavelength, period and speed of the wave (use the 'pause' and 'step' buttons to measure the length of time it takes a peak to pass a given location in the simulation).

Exercise 1.7.4

How does the motion at the top of the water compare with the motion at the bottom?

Exercise 1.7.5

Describe the overall motion of one of the red dots; what path does the dot follow?

Exercise 1.7.6

Earthquakes can produce several different kinds of waves, each traveling at a different speed. Search for a reliable source and find a definition for P-waves, S-waves, Rayleigh waves and Love waves. Be sure to include comments on their speeds and which ones are more destructive.

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1.8: Two-Dimensional Waves

Tutorial 1.8: Two-Dimensional Waves

So far these simulations have only shown one dimensional waves. Even though there is motion perpendicular to the direction the wave travels for a transverse wave, the function describing the wave is a function of only one spacial variable, x . Waves can exist in two or three dimensions, however. One example is a **plane wave** where the wave front or crest of the wave makes a line (in two dimensions) or a plane (in three dimensions). **Circular waves** (in two dimensions) and **spherical waves** (in three dimensions) also exist. The present simulation shows plane and circular waves in two dimensions.

The simulation below starts by showing you a plane wave in two dimensions traveling in the $x - y$ plane, in the x direction, viewed from above. In these simulations the amplitude (in the z direction, towards you) is represented in greyscale. When the wave has a positive amplitude the color is white, when the amplitude is zero the color is light gray and when the amplitude is negative the color is black. The wavelength λ is in centimeters and the period, T is in seconds.

Two-Dimensional Waves

Questions:

Exercise 1.8.1

Click on the 'play' button with the plane wave selected. Experiment with wavelength and period. Is the simulation accurate in representing a fixed speed for a real wave? How do you know? Consult 1.2 on wave speed if you are unsure of your answer.

Exercise 1.8.2

Can this representation be used to describe longitudinal waves as well as transverse waves? Why or why not?

Exercise 1.8.3

The answer to the previous question is 'yes'. What would the white, grey and black colors represent if this was a representation of a longitudinal wave?

Keep in mind the wave shown is really a function of just two variables, x and t . For a wave traveling in the x direction the amplitude is the same for any value of y so the equation describing it is exactly the same as the one dimensional wave we have seen before: $z(x, y, t) = z(x, t) = A \sin(kx - \omega t + \varphi)$. This kind of wave is called a **plane wave**.

The equation for a plane wave traveling in an *arbitrary* direction in the $x - y$ plane is given by $z(x, y, t) = A \sin(k_1 x + k_2 y - \omega t + \varphi)$ where $z(x, y, t)$ is the height of the wave at location (x, y) at time t . Now the wavenumber k has become a *wave vector* with components $k_1 = k \cos \theta$ and $k_2 = k \sin \theta$ where θ is angle between the direction the wave is traveling and the x axis. This equation can also be written using a vector dot product as $z(x, y, t) = A \sin(\mathbf{k} \cdot \mathbf{r} - \omega t + \varphi)$ where $\mathbf{r}$ and $\mathbf{k}$ are vectors with components in the x , y and (for waves moving in three dimensions) z directions. The vector $\mathbf{r}$ gives the direction the wave is traveling and the wave vector $\mathbf{k}$ has components of the wavelength in each direction.

Exercise 1.8.4

For what angle, θ , does the equation of a plane wave in an arbitrary direction become the same as the equation for a wave in the x direction?

Exercise 1.8.5

Now select the circular wave button. Suppose this circular wave simulation represents waves in a pan of water. Given a pan of water and other equipment of your choice in the laboratory, how could you create these waves? Explain the procedure in detail.

Exercise 1.8.6

What is happening to the curvature of the waves as they move away from the source? Is it becoming more or less like a plane wave? Explain why it is reasonable to think of light from the sun as plane waves when the light arrives at the earth, even though they are actually spherical waves moving outwards in all directions from the sun.

Exercise 1.8.7

The circular wave simulation is **unphysical** in one sense because the amplitude of the circular wave does not change as the diameter gets bigger. Why is this unrealistic? Would you expect the amplitude of a circular wave to be the same once it has spread out over a very large distance from the source? Explain.

As noted previously the **Intensity** of a wave is defined to be the power per meter squared and is measured in W/m^2 . Power is energy transmitted per time and the energy of a wave is proportional to the amplitude squared so intensity will also be proportional to amplitude squared.

Exercise 1.8.8

In the simulation, does the intensity of the circular wave change as you get further from the source? How is this different in the case of real waves (for example sound) as you get further from the source?

Many waves spread out in three dimensions from what is essentially a point source, for example a small lamp sends out light in nearly all directions. The surface of a sphere, centered on the point source would catch all of the waves from the source. This imaginary sphere has a surface area of $4\pi r^2$ where r is the distance from the source. So at a distance of r the energy that left the source is spread over area $4\pi r^2$. This means the energy per area is $E/4\pi r^2$ and so decreases as $1/r^2$. We can thus conclude that the intensity of a spherical wave is inversely proportional to distance squared.

Exercise 1.8.9

Suppose the intensity of a spherical wave is 80 W/m^2 at a distance of one meter. What is the intensity at 2 m? 4 m? $1/4 \text{ m}$?

Exercise 1.8.10

If the amplitude squared is proportional to intensity, by what factor does the amplitude of the wave in the previous question change in going from 1 m to 2 m? How much change is there in going from 1 m to $1/4 \text{ m}$?

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CHAPTER OVERVIEW

2: Combining Waves

Some of the most important properties of waves arise due to superposition, the idea that when two waves meet their displacements simply add. These tutorials explore the consequences of superposition.

[2.1: Adding Two Linear Waves \(Superposition\)](#)

[2.2: Interference](#)

[2.3: Group Velocity](#)

[2.4: Other Wave Functions](#)

[2.5: Fourier Analysis and Synthesis](#)

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2.1: Adding Two Linear Waves (Superposition)

Tutorial 2.1: Adding Two Linear Waves (Superposition)

The waves we have been discussing so far and the ones that are most often seen in everyday life, such as light and sound, are for the most part **linear waves**. Linear waves have the property, called **superposition**, that their amplitudes add linearly if they arrive at the same point at the same time. This gives rise to several interesting phenomena in nature which we will investigate in this and the next few simulations.

The simulation shows the function $f(x, t)$ in red, $g(x, t)$ in the blue and $u(x, t) = f(x, t) + g(x, t)$ in grey. Initial wavenumber and angular velocity are equal to one. The check boxes on the lower right determine which functions are visible. You can enter your own functions for $f(x, t)$ and $g(x, t)$ using the same notations used for spreadsheets and calculators.

Wave Superposition

Questions:

Exercise 2.1.1

Uncheck the $g(x, t)$ function so that just $f(x, t)$ is showing. What is the amplitude (half the height from highest point to lowest point) of $f(x, t)$? Now measure the amplitude of $g(x, t)$ (they should be the same). Now find the amplitude of the sum, $u(x, t)$. How does the amplitude of the sum compare with the amplitude of $f(x, t)$ or $g(x, t)$? This is an example of **constructive interference**; the two waves add to give a wave with an amplitude which is the sum of the two amplitudes.

Exercise 2.1.2

Start the simulation (button on lower left). How does the wavelength, frequency and speed of $f(x, t)$ or $g(x, t)$ compare with the wavelength, frequency and speed of $u(x, t)$?

Exercise 2.1.3

Change $f(x, t)$ to have a phase of π (the simulation reads pi as π ; type or cut and paste $2.5 * \sin(x - t + pi)$ for $f(x, t)$ and press return or enter). Run the simulation. What happens to the amplitude of the sum of the two waves, $u(x, t)$? This is an example of **destructive interference**. Write a definition for destructive interference in your own words.

Exercise 2.1.4

Experiment with cases in between total destructive and total constructive interference by changing the phase of $f(x, t)$ to be $1/2\pi$, $1/3\pi$, and $1/4\pi$. Stop the simulation each time and record the amplitude of the sum compared to the amplitude of $f(x, t)$ or $g(x, t)$.

Exercise 2.1.5

Click the reset button (fourth button on lower left) and then change the amplitude of $f(x, t)$ from 2.0 to 3.0 and the amplitude of $g(x, t)$ from 2.0 to 1.0 (Hit Return or Enter to update the values). What is the amplitude of $f(x, t) + g(x, t)$ in this case? How does this amplitude compare to the original case?

Exercise 2.1.6

Go back to the original functions but change one of the minus signs to a plus sign (so now $f(x, t) = 2.0 * \sin(x + t)$ and $g(x, t) = 2.0 * \sin(x - t)$). The sum $u(x, t)$ is called a **standing wave** in this case (an example would be the waves on a guitar string). Describe the behavior of $u(x, t)$. How does the period and wavelength of the combined wave compare to the period and wavelength of two components? How is the maximum amplitude of the sum related to the amplitudes of the two components? What can you say about the speed of the sum?

Exercise 2.1.7

For standing waves on a string a **node** is a location where there is no motion and an **anti-node** is a location where there is maximum motion. For the standing wave in the previous exercise, how many nodes are there? How many anti-nodes?

Exercise 2.1.8

Use trigonometric identities to show that the sum of $f(x, t) = A \sin(kx + \omega t)$ and $g(x, t) = A \sin(kx - \omega t)$ equals $2A \cos(\omega t) \sin(kx)$. We can interpret this as a time dependent amplitude, $2A \cos(\omega t)$, multiplying a sine wave which is fixed in space. What happens to the amplitude as time increases? What fixes the location of the maximums and minimums of the standing wave (Hint: $k = 2\pi/\lambda$)?

Exercise 2.1.9

Notice that the standing wave has zero amplitude on both ends in the simulation. This means that only certain wavelengths will "fit" on a given length. See if you can adjust x min and x max so that you have a wave with more nodes and anti nodes that fits on a longer string with the amplitude still zero on the ends. (Hint: $6.28 = 2\pi$.)

Exercise 2.1.10

Now enter the following functions: $f(x, t) = 2.0 * \sin(x - t)$ and $g(x, t) = 2.0 * \sin(1.1 * x - 1.1 * t)$ (you can cut and paste instead of typing). Watch the just the sum $u(x, t) = f(x, t) + g(x, t)$ for a while and describe what happens (it changes slowly). Two waves with slightly different frequencies added together give rise to the phenomena of **beats**. Now turn $f(x, t)$ and $g(x, t)$ on. Are these waves still traveling at the same speed as $u(x, t)$? Find the beat frequency the following way: Stop the simulation when the two source waves exactly cancel ($f(x, t) + g(x, t)$ is a straight line) and record the time (use the step buttons if you overshoot). Start the simulation and stop it again the next time the waves cancel. Record the new time and subtract to get the elapsed time. The beat frequency is $1/(\text{time elapsed})$. How does this compare to the frequency of $f(x, t)$ subtracted from the frequency of $g(x, t)$?

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